UK Legal Knowledge Base: Strategy and Pipeline (Road Traffic)

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Goal and scope

Build an open-source, accurate knowledge base of UK law that an AI can retrieve from to answer real-life legal questions, with every answer traceable to an official source.

- **First practice area:** road traffic only. Consumer rights and debt, then housing, follow on the same pipeline.
- **Storage:** a single GitHub repository. No separate storage service at this stage.
- **Language:** Python for the whole pipeline.
- **Cost:** zero. Free official APIs, free public repository, free GitHub Actions.
- **Focus:** data accuracy first, then data structure. The LLM and the database layer come later.

Why road traffic first: people ask about it often, the answers are exact and checkable (points, fine bands, time limits), and it is mostly UK-wide, so jurisdiction issues are minimal.

UK legal source hierarchy

Correct answers usually come from levels 2 to 4. The Act often only says where the power comes from.

Level	What it is	Made by	Binding?	Road traffic example	In lexifina?
1. Acts	Primary legislation	Parliament and devolved parliaments	Yes	Road Traffic Act 1988	Yes (4,375 Acts in force)
2. Statutory Instruments	Regulations, Orders, Rules made under an Act	Ministers	Yes	Fixed Penalty Order 2000	Almost none (1 SI)
3. Rules and codes under an Act	Detailed rules that are not formal SIs	Minister or regulator	Usually, or as evidence	Highway Code, Sentencing Council guidelines	No

Level	What it is	Made by	Binding?	Road traffic example	In lexifina?
4. Guidance	How authorities apply the law	Departments, regulators	No	GOV.UK and DVLA pages	No
5. Case law	How courts interpret levels 1 to 4	Judges	Yes (precedent)	High Court appeals	No (none for the UK)

The lexifina/global-law repository covers level 1 only, and some of its texts carry unapplied changes (Road Traffic Act 1988: 295; Road Traffic Offenders Act 1988: 104). Use it as a reference design and a cross-check, not as the source.

Data principles

These six rules decide accuracy; every pipeline step enforces them.

1. **Official primary sources only.** Text comes from the publisher (legislation.gov.uk, GOV.UK, Find Case Law). Law firm articles, Citizens Advice and Shelter are only used privately to write test answers.
2. **Verbatim text, never rewritten.** Summaries and embeddings are derived data, kept apart and regenerated when the source changes.
3. **Every unit traceable.** Each file records its source URL, fetch time, version and SHA-256 hash.
4. **Time is part of the data.** Each version has a valid-from date and, once replaced, a valid-to date. Questions are answered as at a date.
5. **Flag what you can't trust.** Store the unapplied-changes count from legislation.gov.uk and show it in answers.
6. **Clear licences.** Record the licence per document.

Road traffic area scope

The first version targets 40 to 60 units across all five levels, listed in `areas/road-traffic.yaml`.

Level	Sources	Publisher
Acts	Road Traffic Act 1988, Road Traffic Offenders Act 1988, Road Traffic Regulation Act 1984, Road Traffic (New Drivers) Act 1995, Magistrates' Courts Act 1980 (procedure sections), Sentencing Act 2020 (fines sections)	legislation.gov.uk
SIs	Fixed Penalty Order 2000, Traffic Signs Regulations and General Directions 2016, Road Vehicles (Construction and Use) Regulations 1986 (phone use)	legislation.gov.uk
Rules and codes	Highway Code, Magistrates' Court Sentencing Guidelines (speeding, s.172 failure, fine bands)	GOV.UK, Sentencing Council
Guidance	Speeding penalties, penalty points, new drivers, paying a fixed penalty	GOV.UK
Case law	Out of scope for version 1	Find Case Law

Test questions

Each test stores the question, the expected answer and the expected source IDs. Expected answers are confirmed against the sources before they are fixed.

Question	Expected answer	Expected sources
Caught at 41 mph in a 30 zone: how many points?	3 points with a fixed penalty; 4 to 6 points or a short ban if it goes to court	RTOA 1988 Sch 2, speeding guideline
How many points before a new driver loses their licence?	6 within 2 years of passing	Road Traffic (New Drivers) Act 1995
How many points lead to a ban?	12 within 3 years, minimum 6-month ban	RTOA 1988 s.35
Time limit for a notice of intended prosecution?	14 days	RTOA 1988 s.1
Penalty for not naming the driver?	6 points and a fine up to £1,000	RTA 1988 s.172, RTOA 1988 Sch 2

Question	Expected answer	Expected sources
Missed the fixed penalty deadline and got a summons: options?	Respond to the court; statutory declaration within 21 days if unaware of the proceedings	MCA 1980 s.14, RTOA 1988 Part III
Penalty for using a hand-held phone while driving?	6 points and a £200 fixed penalty	Construction and Use Regs reg.110, RTA 1988 s.41D

The full set grows to about 30 questions before the first build is called done.

Unit of storage and identifiers

One file per smallest citable unit, because answers cite "s.172", not "the Road Traffic Act", and smaller units make retrieval more precise. Always use the publisher's identifier where one exists.

Source type	Unit	Stable ID
Acts and SIs	Section or regulation, subsections inside	`ukpga/1988/52/section/172`
Highway Code	Rule	`highway-code/rule/124`
Sentencing guidelines	One offence guideline	`sentencing/magistrates/speeding`
Guidance	Page section	`govuk/<path>#<heading>`
Case law (later)	Judgment, split into paragraphs	Neutral citation + paragraph

Explanatory notes are their own units: one file per note section, `level: explanatory-note`, `binding: guidance`, linked to the section they explain through `relations.cites`. They come from legislation.gov.uk under the same licence and explain in plain English what each section is for.

Metadata schema

Each file in `data/` is Markdown with YAML front matter; `schema/provision.schema.json` validates the front matter and the verbatim legal text sits below it. Three fields matter most: `level` and `binding` (law or guidance), `version` (answers as at a date) and `extent` (where it applies).

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "Legal provision",
  "type": "object",
  "required": ["id", "title", "level", "binding", "extent", "areas",
"version", "source"],
  "properties": {
    "id": { "type": "string" },
    "title": { "type": "string" },
    "level": { "enum": ["act", "si", "rule", "code", "guideline", "guidance",
"explanatory-note", "caselaw"] },
    "binding": { "enum": ["binding", "evidential", "persuasive", "guidance"]
},
    "parent": { "type": ["string", "null"] },
    "made_under": { "type": ["string", "null"] },
    "extent": { "type": "array", "items": { "enum": ["E", "W", "S", "NI"] },
"minItems": 1 },
    "areas": { "type": "array", "items": { "type": "string" }, "minItems": 1
},
    "version": {
      "type": "object",
      "required": ["valid_from", "status"],
      "properties": {
        "valid_from": { "type": "string", "format": "date" },
        "valid_to": { "type": ["string", "null"], "format": "date" },
        "status": { "enum": ["in-force", "prospective", "repealed"] }
      }
    },
    "unapplied_changes": { "type": "integer", "minimum": 0 },
    "relations": {
      "type": "object",
      "properties": {
        "amended_by": { "type": "array", "items": { "type": "string" } },
        "cites": { "type": "array", "items": { "type": "string" } },
        "interprets": { "type": "array", "items": { "type": "string" } }
      }
    },
    "source": {
      "type": "object",
      "required": ["url", "publisher", "retrieved_at", "sha256", "licence"],
      "properties": {
        "url": { "type": "string", "format": "uri" },
        "publisher": { "type": "string" },
        "format": { "type": "string" },
        "retrieved_at": { "type": "string", "format": "date-time" },
        "sha256": { "type": "string", "pattern": "^[a-f0-9]{64}$" },
```

```
        "licence": { "type": "string" }
      }
    }
  }
}
```

Example file, `data/legislation/ukpga/1988/52/section-172/current.md`:

```
---
id: ukpga/1988/52/section/172
title: "Duty to give information as to identity of driver etc in certain
circumstances"
level: act
binding: binding
parent: ukpga/1988/52
made_under: null
extent: [E, W, S]
areas: [road-traffic, criminal]
version:
  valid_from: 2015-04-13
  valid_to: null
  status: in-force
unapplied_changes: 2
relations:
  cites: [ukpga/1988/53/section/1]
source:
  url: https://www.legislation.gov.uk/ukpga/1988/52/section/172/2015-04-13
  publisher: legislation.gov.uk
  format: CLML XML
  retrieved_at: 2026-11-09T10:00:00Z
  sha256: <64-character hash>
  licence: OGL-3.0
---

(2) Where the driver of a vehicle is alleged to be guilty of an offence...
```

The values in the example are illustrative; real values come from the fetched XML.

Repository structure

Folders are organised by source and tagged by practice area, because one provision often belongs to several areas (RTA s.172 is both traffic and criminal). Everything lives in one GitHub repository.

```
uk-law-kb/
├── README.md
├── SOURCES.md                  source, licence, fetch method, refresh
frequency
├── schema/
│   └── provision.schema.json
├── raw/                        exact API responses, never edited
│   └── legislation/ukpga/1988/52/section-172/2015-04-13.xml
├── data/                       normalised Markdown + front matter
│   ├── legislation/ukpga/1988/52/section-172/
│   │   ├── current.md
│   │   └── versions/2015-04-13.md
│   ├── codes/highway-code/rule-124.md
│   ├── guidelines/sentencing/magistrates/speeding.md
│   └── guidance/govuk/...
├── areas/
│   └── road-traffic.yaml       units in the area, key concepts
├── tests/
│   └── road-traffic/questions.yaml
└── tools/
    ├── fetch/
    ├── normalise/
    ├── validate/
    └── evaluate/
```

Folder	Role	Rebuildable?
`raw/`	Audit trail and backup: what the publisher said, when. One file per version, named by date; identical re-fetches are not saved.	No
`data/`	What the AI reads to answer	Yes, from `raw/`
`areas/`	The list of units to fetch per practice area	Hand-written
`tests/`	Questions with expected answers and expected source IDs	Hand-written
Search index	Embeddings for retrieval, kept outside git	Yes, from `data/`

Expected size for road traffic with version history: a few MB; three areas stay under about 50 MB.

Pipeline

The pipeline runs in Python, in this order: fetch, normalise, validate, review, evaluate. GitHub Actions runs it weekly and opens a pull request when anything changes.

Step	What it does	Python tools
1. Fetch	Reads `areas/road-traffic.yaml`, calls the official APIs, saves exact responses to `raw/` with a hash	`httpx`
2. Normalise	Converts XML or JSON into Markdown + front matter in `data/`; closes the old version with `valid_to` when a new one appears	`lxml`, `pyyaml`
3. Validate	Schema check, hash check, every relation points to an existing ID, `valid_from` before `valid_to`	`jsonschema`
4. Review	A person checks anything the parser can't handle with confidence (tables, schedules)	Pull request review
5. Evaluate	Runs the test questions; checks answers and that retrieval found the expected source IDs	Own script

Rules:

- A wrong parse is worse than a missing one: when unsure, flag for review instead of publishing.
- A drop in the test score fails the build.
- `raw/` is never edited. A parser fix means rebuilding `data/` from `raw/`.
- Copy and adapt the CLML XML parser from [i-dot-ai/lex](i-dot-ai/lex) (MIT licence, keep its copyright notice) into our normaliser. It already maps extent (E, W, S, NI). We copy the code only and never depend on the lex hosted API, which is experimental and unversioned.

Fetching the data

All road traffic sources are free and need no API key.

Source	How to fetch	Example
legislation.gov.uk (Acts, SIs)	Add `/data.xml` to any URL; add a date for a past version	`https://www.legislation.gov.uk/ukpga/1988/52/section/172/data.xml`

Source	How to fetch	Example
legislation.gov.uk, past version	Date after the section	`.../section/172/2015-04-13/data.xml`
legislation.gov.uk, new laws	Atom feed	`https://www.legislation.gov.uk/new/data.feed`
GOV.UK (Highway Code, guidance)	Content API: insert `/api/content` after the domain	`https://www.gov.uk/api/content/guidance/the-highway-code`
GOV.UK search	Search API	`https://www.gov.uk/api/search.json?q=speeding`
Sentencing Council	No API: HTML pages, fetched slowly	sentencingcouncil.org.uk

The legislation XML (CLML) already holds extent, version dates and unapplied changes, which fills most of the schema. Wait 5 seconds between legislation.gov.uk requests, as its robots.txt asks, and send a User-Agent with contact details.

```
import hashlib, time, httpx
from pathlib import Path

HEADERS = {"User-Agent": "uk-law-kb (contact: your-email@example.com)"}

def fetch(uri: str, date: str | None = None) -> Path:
    url = f"https://www.legislation.gov.uk/{uri}" + (f"/{date}" if date else
"") + "/data.xml"
    r = httpx.get(url, headers=HEADERS, timeout=30, follow_redirects=True)
    r.raise_for_status()
    out = Path("raw/legislation") / uri / f"{date or 'current'}.xml"
    out.parent.mkdir(parents=True, exist_ok=True)
    out.write_bytes(r.content)
    print(uri, hashlib.sha256(r.content).hexdigest())
    time.sleep(5)  # respect the crawl delay
    return out

fetch("ukpga/1988/52/section/172")
```

Licences

The repository uses two licences: MIT or CC0 for code and schema, and each source's own licence recorded in every file's front matter.

Source	Licence	Republish the text?
legislation.gov.uk	Open Government Licence v3	Yes, with attribution
GOV.UK guidance, Highway Code	Open Government Licence v3	Yes, with attribution
Sentencing Council guidelines	Check their terms	Verify before publishing
Find Case Law (later)	Open Justice Licence	Bulk computational analysis needs a free licence application first
Citizens Advice, Shelter, law firms	Copyright	No: link only, or use privately for test answers

Cost

The knowledge base costs nothing to build and maintain. The only possible cost is the LLM that writes answers, decided later.

Item	Cost
legislation.gov.uk and GOV.UK APIs	Free
Public GitHub repository	Free
GitHub Actions weekly refresh	Free for public repositories
Storage (a few MB)	Free, no Git LFS needed
Embeddings and index (when added)	Free if run locally
Answering LLM (when added)	Free with a local model via Ollama, or paid for a hosted API

Later, out of scope for now

- **Vector DB + graph DB.** The vector DB holds embeddings of `data/` for semantic search. The graph DB holds the `relations` from the front matter (made under, amends, cites, interprets), so answers can follow a rule back to its Act or a section to the cases that interpreted it. Both are built from `data/`, so the schema above already prepares for them.
- **More areas.** Consumer rights and debt next, then housing (England first), which tests date handling because section 21 evictions ended on 1 May 2026.
- **Case law.** Find Case Law, after the licence application.
- **Larger storage.** If `raw/` grows to many GB, move it to Git LFS or a storage bucket and keep `data/` in git.

Next steps

- [ ] Create the GitHub repository with the folder structure and licences
- [ ] Write `schema/provision.schema.json`
- [ ] Write `areas/road-traffic.yaml`: the 40 to 60 units to fetch
- [ ] Write `tests/road-traffic/questions.yaml`: about 30 questions, answers confirmed against sources
- [ ] Build the legislation.gov.uk fetcher; copy and adapt the lex CLML parser into the normaliser
- [ ] Build the GOV.UK fetcher for the Highway Code and guidance
- [ ] Check Sentencing Council reuse terms
- [ ] Add validation and the weekly GitHub Actions refresh
- [ ] Fetch explanatory notes for the road traffic Acts that have them

Note: data accuracy risks to manage

These are the known ways the data can be wrong. Each needs a check in the pipeline or a flag in the answer before the knowledge base is trusted.

Risk	Why it matters	How we handle it
Unapplied changes on legislation.gov.uk	Some texts don't yet show later amendments (RTA 1988 had 295 pending)	Store `unapplied_changes`; warn in answers; cross-check against the amendments data and lexifina

Risk	Why it matters	How we handle it
Prospective provisions	Text is published but not yet in force	`status: prospective`; never answered as current law
Partial commencement	A section can be in force for some purposes or areas only	Record commencement notes from the XML; flag for review
Extent vs application	Where a law extends is not always where it applies; Northern Ireland has its own road traffic laws	Tag `extent`; default to England and Wales and say so in answers
Point-in-time dates	The law at the date of the event decides the answer, not today's law	Keep every version with `valid_from` / `valid_to`; ask or infer the event date
Tables and schedules	RTOA 1988 Schedule 2 (offences and points) is a table; a bad parse gives wrong points	Parse tables as tables; human review of every schedule; tests on points answers
Amendment annotations	legislation.gov.uk text carries brackets and F-notes marking changes	Handle consistently: keep the clean current wording, store annotations as metadata
Explanatory notes date	Notes describe the Act as passed, not as amended, and only exist for Acts from 1999	`binding: guidance`; show the note's date; never override the current section text
Guidance lag	GOV.UK pages can lag behind a change in the law	Law outranks guidance in answers; flag conflicts for review
Guidance and code dates	GOV.UK `public_updated_at` is an edit date, not the legal effective date	Record the effective date from the page or the change notice where stated
Sentencing guideline dates	Guidelines apply from a set date, sometimes to cases sentenced after it	Store the effective date per guideline version
Broken cross-references	A section citing another that we haven't fetched	Validation fails on a missing ID; fetch it or mark it external
Completeness	A unit listed in `areas/road-traffic.yaml` that failed to fetch	Validation checks every listed unit exists in `data/`

Risk	Why it matters	How we handle it
Silent source changes	The publisher edits text without a new version date	Re-fetch weekly; a new hash with the same date is flagged for review
Wrong expected answers in tests	Tests are only as good as their answers	Every expected answer cites its source; ideally reviewed by a solicitor before release